

TQF Quant Pathway Handbook

A structured handbook based on the official TQF Quant Pathway page

Source page: <https://www.tqf.or.th/quant-pathway>

This handbook reorganizes the official TQF Quant Pathway into a clean reference format for self-study, curriculum planning, and capability mapping.

The pathway is organized into three levels: foundational, core, and specialized. Together they describe a practical learning structure for quantitative finance and financial engineering.

Level	Description
Foundational	Essential foundations in finance, mathematics, and programming for anyone entering the quant field.
Core	Cross-functional knowledge that helps quant professionals understand the broader financial industry and
Specialized	Focused areas for deeper expertise in portfolio management, trading, derivatives, credit, and regulation.

Foundational

Finance

- Time value of money (discounting, compounding, annuities)
- Financial instruments (equities, bonds, derivatives basics)
- Portfolio theory basics (CAPM, mean-variance, APT)
- Corporate finance basics (capital structure, leverage, WACC)
- Financial statements (balance sheet, P&L, cash flow)

Mathematics

- Calculus (differentiation, integration, multivariable calculus)
- Linear algebra (matrices, eigenvalues, SVD, optimization foundations)
- Probability and statistics (distributions, estimators, hypothesis testing, regression)
- Differential equations (ODEs, PDEs)

Programming

- Data Handling and Processing
- Statistical and Mathematical Libraries
- Object-Oriented Programming
- Numerical Methods
- Languages: Python, C++, R, VBA

Core

Advanced Mathematical Methods

- Stochastic calculus: Stochastic processes, Markov processes, martingales, Ito's Lemma, change of numeraire
- Statistical methods: Bayesian inference, copulas, and extreme value theory
- Numerical methods: Finite difference methods, Monte Carlo simulations, and variance reduction

Asset Behavior & Modeling

- Asset pricing (no-arbitrage, risk-neutral valuation, CAPM extensions)
- Option pricing theory (Binomial models, Black-Scholes, Greeks, implied volatility)
- Volatility modeling (ARCH/GARCH, stochastic volatility, SABR, volatility skews/smiles, realized volatility)
- Market microstructure basics (order book dynamics, liquidity, transaction costs, market impact models)

Fixed Income and Credit

- Term structure modeling (Vasicek, CIR, Hull-White, HJM, LMM, Nelson-Siegel)
- Yield curve construction and analysis (bootstrapping, interpolation techniques)
- Duration and convexity (key rate duration, partial durations)
- Credit risk models (structural vs reduced-form, Merton model, KMV)
- IFRS 9 / Basel frameworks (PD, LGD, EAD, ECL, stress testing)
- Securitization and structured products (ABS, MBS, CDOs, CLOs)

Risk Management

- Value-at-Risk (parametric, historical, Monte Carlo, tail VaR)
- Expected Shortfall (coherent risk measures)
- Hedging strategies (delta, gamma, vega, cross-asset hedging)
- Liquidity risk, counterparty risk, model risk (model validation, backtesting)
- Stress testing & scenario analysis (extreme events, reverse stress testing)

Machine Learning & Data Science

- Regression, classification, clustering
- Dimensionality reduction (PCA, SVD)
- Time series forecasting
- Feature engineering for financial data
- NLP (sentiment analysis, regulatory texts)
- Reinforcement learning in trading strategies

Specialized

Portfolio Management

- Factor models (Fama-French, risk premia, multi-factor models)
- Portfolio optimization (mean-variance, robust optimization, Black-Litterman model)
- Efficient frontier and Sharpe ratio (risk-adjusted performance metrics)
- Risk parity, smart beta, asset allocation (dynamic asset allocation, ESG investing)
- Performance attribution (Brinson model, multi-period attribution)
- Portfolio stress testing (scenario-based, factor-based, dynamic hedging strategies)

Trading & Execution

- Algorithmic trading (execution algos, TCA, market impact models)
- Market-making models (inventory management, bid-ask spread optimization)
- High-frequency trading (latency, order book modeling, co-location)
- Slippage and market impact (VWAP, TWAP, implementation shortfall)
- Statistical arbitrage and pairs trading

Derivatives and Advanced Products

- XVA (CVA, DVA, FVA, KVA, MVA, regulatory capital adjustments)
- Exotic options pricing (barrier, Asian, quanto, lookback, digital options)
- Real options & commodities (energy markets, real asset valuation)
- Local and stochastic volatility models (Dupire, Heston, SABR extensions)
- Interest rate and currency derivatives (swaps, caps, floors, cross-currency swaps)
- Hybrid derivatives (e.g., equity-credit hybrids)
- Weather and catastrophe derivatives

Credit Risk Management

- Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD)
- Credit portfolio modeling and forecasting
- Credit scoring/rating and early warning system (Machine learning applications)
- Counterparty risk and credit valuation adjustment (CVA)
- Stress testing for credit portfolios under regulatory frameworks

Regulation & Implementation

- Basel III calculations, scenario analysis, capital optimization
- IFRS-related, e.g. expected credit loss, hedge accounting, fair value

- Collateral management (CSA, ISDA, margining, collateral optimization)
- Treasury and ALM models (liquidity coverage ratio, net stable funding ratio)
- Regulatory capital optimization (RWA, leverage ratio, stress testing frameworks)